

Project Design Phase-II

Data Flow Diagram & User Stories

Date	30 September 2026
Team ID	SWTID-2026-9682
Project Name	AI StudyBuddy – AI-Augmented Backend Development
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Level 0 – Context Diagram

The whole backend is shown as one process that exchanges data with the Student, the Administrator and the external Google Gemini API.

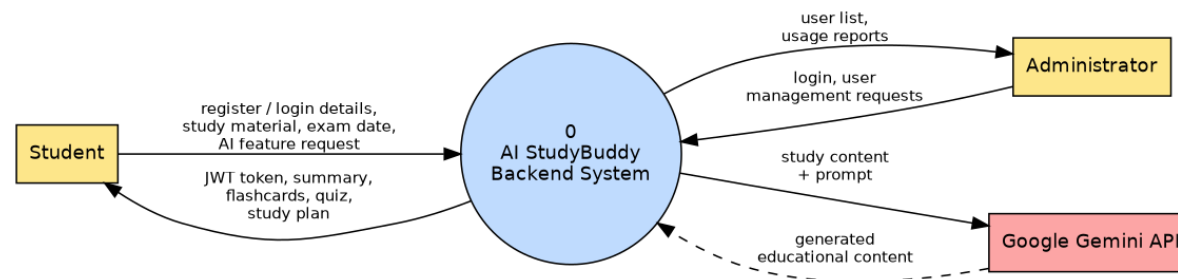


Figure 1: Level 0 DFD of AI StudyBuddy

Level 1 DFD

The system is broken into five processes with three data stores (D1–D3), yellow and red rectangles are external entities.

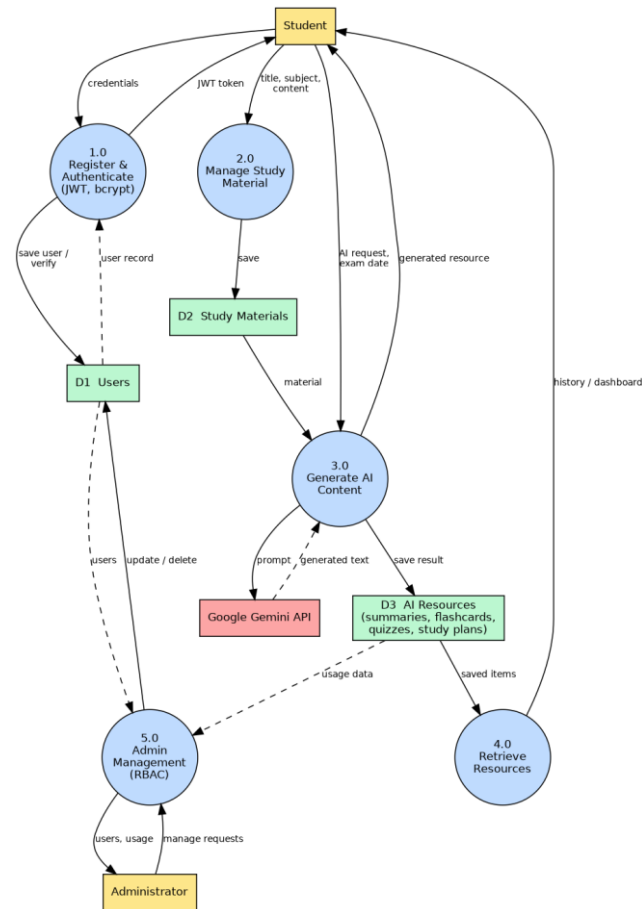


Figure 2: Level 1 DFD of AI StudyBuddy

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Student	Registration	USN-3	As a student, I can register by entering my name, email and password.	Account is created and I can log in	High	Sprint-1
		USN-4	As a student, my password is encrypted before it is stored.	Password in the database is a bcrypt hash, never plain text	High	Sprint-1
	Login	USN-5	As a student, I can log in with my email and password.	I receive a JWT token and reach my dashboard	High	Sprint-1
		USN-6	As a student, only authenticated requests can reach protected features.	Requests without a valid token get 401 Unauthorized	High	Sprint-1
	Study Material	USN-7	As a student, I can upload or enter study material.	Material with title, subject and content is saved and listed	High	Sprint-1
		USN-8	As a student, I can view and delete my saved materials.	Only my materials are shown; deleted items disappear	Medium	Sprint-1
	AI Summary	USN-10	As a student, I can generate a concise AI summary from my material.	A summary is displayed within a reasonable time	High	Sprint-2
		USN-11	As a student, my generated summary is saved.	Summary is available again after I log out and back in	High	Sprint-2
	Flashcards	USN-12	As a student, I can generate flashcards from my material.	Question–answer flashcards are shown and saved	High	Sprint-2
		USN-13	As a student, I can view my saved flashcards.	Saved flashcards load for the selected material	Medium	Sprint-2
	Quiz	USN-14	As a student, I can generate a multiple-choice quiz.	Questions with options and correct answers are generated	High	Sprint-2
		USN-15	As a student, my quiz is saved so I can retake it.	Quiz can be reopened from my dashboard	Medium	Sprint-2
	Study Plan	USN-16	As a student, I can enter my exam date and available time to get a study plan.	A plan covering the days up to the exam is generated	High	Sprint-3

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
		USN-17	As a student, I can view and save my study plans.	Saved plans are listed in my dashboard	Medium	Sprint-3
	Dashboard	USN-18	As a student, I can see all my generated resources in one place.	Summaries, flashcards, quizzes and plans are listed	Medium	Sprint-3
		USN-19	As a student, I can use a React dashboard to upload material and access AI features.	All features are reachable from the dashboard	Medium	Sprint-3
Administrator	Role-Based Access	USN-20	As an admin, I can use admin-only APIs that students cannot access.	Student requests to admin APIs return 403 Forbidden	High	Sprint-3
	User Management	USN-21	As an admin, I can view and manage registered users.	User list is shown and account changes are saved	Medium	Sprint-3
	Monitoring	USN-22	As an admin, I can monitor AI usage and review system logs.	Usage and log details are visible to admin only	Medium	Sprint-4

Note: USN numbers match the Sprint backlog in the Project Planning document. Developer, testing and deployment tasks appear there and not in this student/admin list.

References:

Visual Paradigm – DFD with multiple levels: <https://www.visual-paradigm.com/tutorials/leveled-dfd.jsp>

Node.js Documentation: <https://nodejs.org/en/docs>

Express.js Routing Guide: <https://expressjs.com/en/guide/routing.html>

MongoDB Documentation: <https://www.mongodb.com/docs/>

Mongoose ODM Documentation: <https://mongoosejs.com/docs/>

JSON Web Token Introduction: <https://jwt.io/introduction>

Google Gemini API Documentation: <https://ai.google.dev/gemini-api/docs>

React Documentation: <https://react.dev/>